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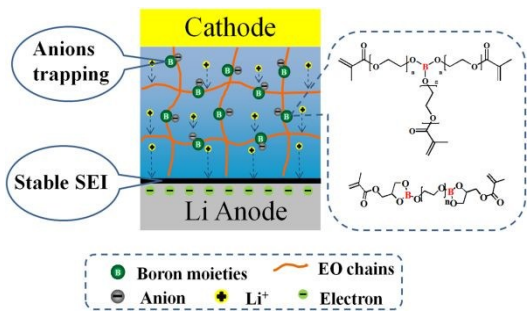
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A borate concentrated polymer electrolyte with near-single ion conduction and stable solid electrolyte interphase for high performance lithium metal batteries



ARTICLE

A Borate Concentrated, Cross-linked Gel Polymer Electrolyte with Near-Single Ion Conduction for Lithium Metal Batteries

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Lithium metal batteries (LMBs) show tremendous potential in energy storage due to ultrahigh energy densities. Nevertheless, maintaining a stable solid electrolyte interphase (SEI) and suppressing the growth of dendrite are hard to achieve with conventional liquid electrolytes. Gel/solid polymer electrolytes are considered sought-after candidates to block the dendrite growth, while they are still suffering from poor ionic conductivity and incompatible interphase. Herein, a borate-concentrated gel polymer electrolyte with 3D cross-linked structure (3D-BGPE) is designed to settle these problems. By incorporating anion-trapping boron moieties into 3D configuration via in-situ strategy, the 3D-BGPE presents high ionic conductivity (8.4×10^{-4} S/cm), near single ion conduction (Li^+ transference number 0.76) and construct a stable and conductive SEI layer on lithium anode. Accordingly, the LMBs employing 3D-BGPE can effectively inhibit the dendrite growth and maintain 89.73% capacity retention after 400 cycles. A promising approach to design high effective electrolytes for safe and long-life LMBs was provided in this work.

1. Introduction

Lithium metal is regarded as a desired anode material because of ultrahigh theoretical specific energy (3860 mAh/g) and low electrochemical potential (-3.04 V vs standard hydrogen electrode).^{1, 2} Nevertheless, the development of lithium metal batteries (LMBs) with liquid electrolytes is hindered by the uneven lithium deposition and dendrite growth during charge/discharge cycles that cause short lifetime, low Coulombic efficiency, and security issues including short circuits and thermal runaway.³ Given the safety hazards of electrolyte leakage and the inherent instability between lithium metal and liquid electrolytes (LEs), the substitution of liquid electrolytes with gel/solid electrolytes provides opportunities to build safer LMBs.^{4, 5}

Over these years, the fast development of solid electrolytes provides an essential solution to tackle lithium dendrite growth in LMBs.¹ Although inorganic solid electrolytes could efficiently suppress lithium dendrites, however, their brittleness and interface contacting issues between lithium

metal anode and electrolyte restrict their further applications.⁶ In comparison, polymer electrolytes with high mechanical flexibility mitigate the contact problems, but the low ion conductivity at room temperature greatly imposed the constraints on their applications.⁷ Therefore, with the purpose of addressing the mentioned problems, gel polymer electrolytes (GPEs) with superior ion conductivity are expected to be an appropriate candidate to realize the dendrite-free Li deposition by maintaining a stable SEI film on lithium anode.^{8, 9} However, GPEs usually suffered from low mechanical strength and poor Li^+ transference number (t_{Li^+}).¹⁰

Several approaches have been developed to enhance Li^+ transference number by introducing single-ion conductor,^{11, 12} porous materials (MOFs),¹³ nano fillers such as Al_2O_3 ,^{14, 15} and SiO_2 ,¹⁶⁻¹⁹ TiO_2 ,²⁰ and Lewis acids to the electrolytes. Among these strategies, introducing boron into the polymer matrix is an attractive method. With a vacant p-orbital, boron work as a Lewis acidic center that interacts with the anion of electrolyte salts to increase the dissociation of ion pairs and improve ionic conductivity and transfer number. For instance, boron moieties were introduced into the polymer matrixes,²¹⁻²³ the inorganic fillers,^{24, 25} and the electrolyte salts²⁶ of GPEs via different approaches to realize the improvements on the electrochemical properties. Nevertheless, there have been few discussions about the ability of boron-based anion receptors for facilitating the formation of stable SEI film to regulate the uniform lithium stripping/deposition. Furthermore, most GPEs were fabricated by blending lithium salts with linear polymer matrixes, the resulting GPEs generally possess low mechanical

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strength and poor dimensional stability, and fail to suppress lithium dendrite growth.

Here, we report a novel 3D-BGPE rich in boron ester with good mechanical strength, high Li^+ transference number, and prominent ionic conductivity. Polymerized by two kinds of cross-linkers, the 3D-BGPE simultaneously introduces cyclic boron ester and branched boron ester in the 3D structure, and thereby enhance the thermo-stabilization and improve the ion transference. Thus, the introduction of anion-trapping boron moieties improves the ion conductivity (with near single ion conduction) and improves the stability of SEI layer. With these benefits, the 3D-BGPE offers efficient lithium dendrites blocking, with 89.73% retention after 400 cycles in the $\text{LiFePO}_4/\text{3D-BGPE}/\text{Li}$ cells, which sheds light on the electrolyte development for energy storage devices.

2. Experimental Section

2.1 Chemicals and Materials

Poly(ethylene glycol) methacrylate (PEGMA, average $M_n = 360$ g/mol), Poly(ethylene glycol) (PEG, average $M_n = 400$ g/mol) trimethyl borate (TMB, $\geq 99.8\%$), Acetonitrile (CH_3CN , 99.0%), were obtained from Aladdin. Glycerol mono-methacrylate (GMMA, 97%) was purchased from J&K Scientific. 1.0 mol/L lithium bis(trifluoromethane sulfonyl) imide (LiTFSI) dissolved in a mixture of ethylene carbonate (EC) and dimethyl carbonate (DMC) (1:1 vol%) solution purchased from Aladdin.

2.2 Synthesis of Boron-Containing Cross-Linkers

2.2.1 Synthesis of linear boron-containing cross-linker (LBC)

TMB (4.16 g, 0.04 mol) and GMMA (6.40 g, 0.04 mol) were mixed in 30 mL anhydrous acetonitrile (CH_3CN) and the solution was stirred at 65 °C for 2 h in dry N_2 atmosphere. Then, PEG (8 g, 0.02 mol) was injected to the solution and continuously react 4 h at 70 °C. The organic solvent was eliminated by rotary evaporation and the product was dried under vacuum at 70 °C over 12 h. After the operation, the LBC was obtained with almost 89.73% yield and stored in an argon filled glove box.

2.2.2 Synthesis of branched boron-containing cross-linkers (TBC)

TMB (0.426 g, 0.0041 mol) and PEGMA (4.32 g, 0.012 mol) were mixed in 30 mL anhydrous acetonitrile (CH_3CN) and the solution was stirred at 60 °C for 4 h in N_2 atmosphere. Then, the temperature increased to 70 °C and continuously stirred for 3 h under N_2 purging to remove methanol (CH_3OH) that produced as a byproduct. Residual ingredients and solvent were evaporated under reduced pressure at 60 °C and dried under vacuum at room temperature over 48 h to get 4.4 g of TBC with 87% yield and stored in an argon filled glove box.

2.3 Synthesis of Gel Polymer Electrolytes

Gel polymer electrolytes (GPEs) were prepared through in situ thermal polymerization method. Stoichiometric amounts of the TBC and LBC were mixed into a bottle as precursors at different molar ratios. Then 1 mol/L LiTFSI and 0.5 wt% AIBN were added into the nonaqueous solution (EC:DMC = 1:1) with stirring over 20 min until all the components were

monodispersed. Mixed solution was injected into a 2016-type coin cell, where a cellulose membrane was used as the supporting medium for the 3D-BGPEs. Followed by the batteries were heated under 70 °C for 1 h. All steps for producing the 3D-BGPEs were handled in an argon filled glove box.

2.4 Characterization

Structural information of the cross-linkers was investigated by ^1H , ^{11}B , ^{13}C nuclear magnetic resonance (NMR) spectroscopy (Avance III 400 MHz Digital NMR spectrometer). Fourier transfer infrared spectroscopy (FT-IR) tests were carried out on an FT-IR spectrometer (Nicolet 6700). The morphologies of 3D-BGPEs and Li metal foil were characterized by scanning electron microscopy (SEM, Nova Nano SEM 230). The powder X-ray diffraction (XRD) measurements with $\text{Cu K}\alpha$ radiation. Thermogravimetric analysis (TGA) was performed at a heating rate of 10 °C/min under N_2 atmosphere. X-ray photoelectron spectroscopy (XPS) analysis was conducted on an ESCALAB 250Xi system (Thermo-Fisher Scientific). Stress-strain curves were investigated using a universal mechanical analysis (DMA Q800, TA instruments) under a rate of 1% elongation per minute.

2.5 Electrochemical Analysis

Ionic conductivity of the 3D-BGPE was measured by electrochemical impedance spectroscopy (EIS) using a PARSTAT4000 electrochemical potentiostat/galvanostat system. A potential of 10 mV was implemented to stainless steel (SS) electrode symmetrical cells with frequency ranging from 0.1 Hz to 1 MHz. The ionic conductivity can be calculated by the following equation:

$$\sigma = l / (R_b A)$$

where l stands for the film thickness, R_b represents the bulk resistance, and A represents the area.

The Li^+ transference numbers (t_{Li^+}) were measured by DC polarization/AC impedance in symmetric Li/3D-BGPE/Li cells. The t_{Li^+} value calculated using the following equation:

$$t_{\text{Li}^+} = I_{\text{ss}}(V - I_0 R_0) / [I_0(V - I_{\text{ss}} R_{\text{ss}})]$$

where V is the polarization voltage applied, I_0 and R_0 represent the initial current and interfacial resistance correspondingly, and I_{ss} and R_{ss} are the steady state current and interfacial resistance, respectively.

The electrochemical stability of the 3D-BGPE investigated by cyclic voltammetry scan with SS/3D-BGPE/Li coin cells. The negative scan was swept from 2.5 V to -0.5 V and then back to 2.5 V and the positive scan was from 2.5 V to 5 V at a rate of 5 mV/s.

2.6 Battery Assembly and Measurement

For the cathode, LiFePO_4 , carbon black, and PVDF were mixed in an 80:10:10 weight ratio in N-methyl-2-pyrrolidone (NMP). The mixture was fully stirred over 2 h to form a slurry. The slurry was casted onto Al foil and dried in a vacuum oven at 110 °C for 12 h to remove the NMP. The mass loading of active material was about 1.0 mg/cm². All $\text{LiFePO}_4/\text{3D-BGPE}/\text{Li}$ coin cells were assembled in an argon-filled glove box. Galvanostatic charge-discharge tests were carried out on LANHE CT2001A systems with the potential range of 2.5-4.0 V.

2.7 Calculation Methods

All the DFT calculations were performed with Yang-Parr correlation functional (B3LYP) utilizing the framework of Gaussian 09 program package. To simplify our calculations, long EO chains were replaced by three-repeating-unit groups, and this simplified method had been reported previously.^{27, 28} Geometry optimizations were performed with a 6-31G* basis set, while the reported single-point energies have been obtained with a larger 6-31+G(d,p) basis set. For enhancing the accuracy of the weak interaction energy between molecules, the basis set superposition error (BSSE) calculations were carried out through interaction energy calculations with the 6-311+G(d,p) basis set.

3. Results and Discussion

3.1 Structural Characterization

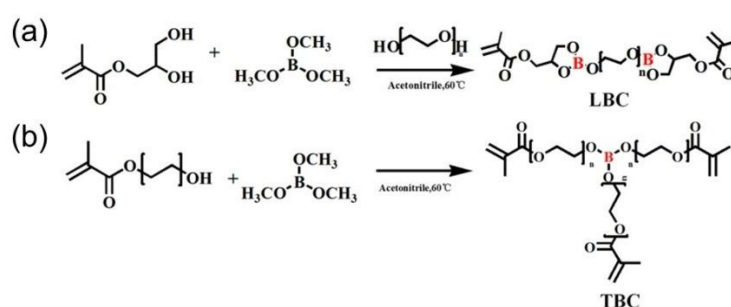
In this work, we designed and synthesized two kinds of boron-containing cross-linkers through a mild transesterification reaction. The linear boron-containing cross-linker (LBC) consists of two cyclic boron ester moieties and poly(ethylene glycol) (PEG) bridges, as presented in Scheme 1(a). The branched boron-containing cross-linkers (TBC) with boron atom as a centre, three poly(ethylene glycol) methacrylate (PEGMA) molecules constitute a Y-shape structure was synthesized via specific procedure of esterification, as shown in Scheme 1(b).

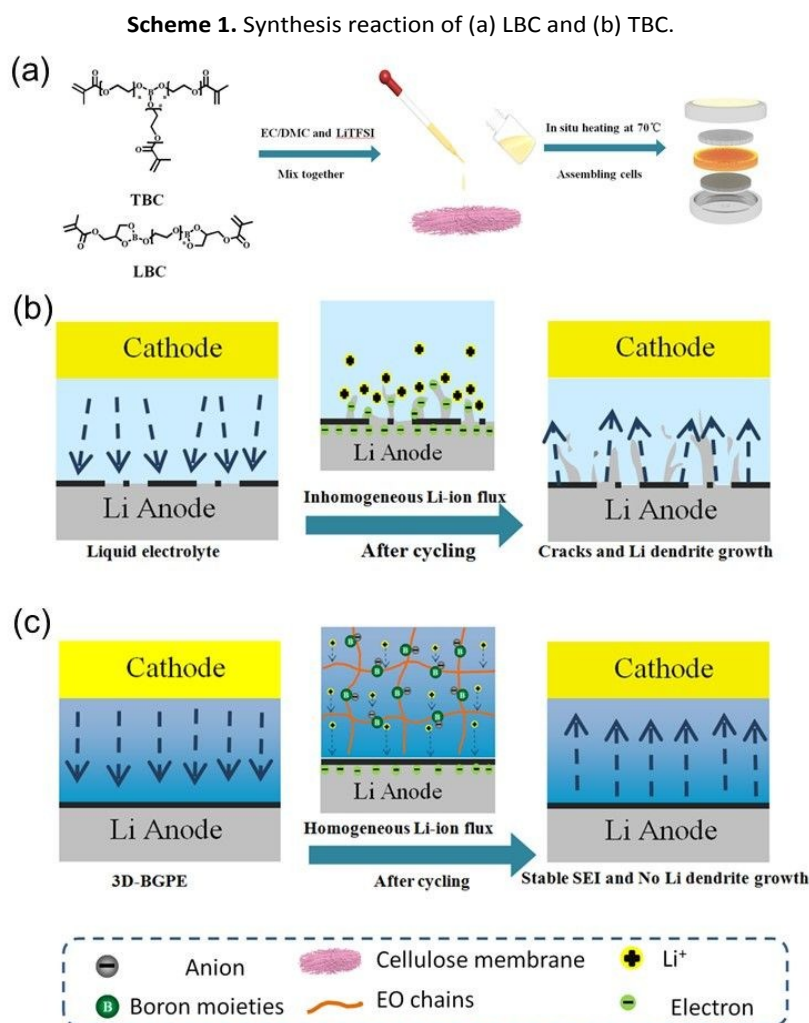
The cross-linked 3D-BGPE was constructed by the LBC and TBC through thermal initiated radical polymerization. Cellulose membrane is consisted of randomly arranged nanofibers with high porosity which offer framework for chain movement and also act as a supporting medium for the polymer electrolytes. The synthetic procedure is illustrated in detail in Scheme 2(a). The LBC and TBC were mixed with a certain molar ratio (LBC:TBC=10:1 is the optimized ratio according to experiments). As illustrated in Scheme 2(b), compared with traditional liquid electrolytes where anions move freely and nonuniform lithium deposition, the crosslinked 3D-BGPE with anion trapping ability enables a small concentration gradient of Li⁺ and uniform Li-ion distribution, and in turn, results in the suppression of lithium dendrite. Notably, the in situ polymerization simplifies the manufacturing procedure of the LMBs and contributes to improved contacting property of electrolyte and anode compared with traditional solid electrolytes.^{29, 30}

¹H-NMR analysis verifies the successful synthesis of LBC. Figure 1(a) exhibits the ¹H-NMR spectra of the reactants (PEG and GMMA) and the product LBC. Compared with ¹H-NMR spectra of (PEG and GMMA) where the signals completely in accordance with their structure, the formation of boron ester structure was verified by the disappearance of the hydroxyl signals at 4.56 ppm (PEG, peak h), 4.65 ppm (GMMA, peak i). Overlapping peaks observed at 6.03 ppm (a), 5.68 ppm (b) and 1.89 ppm (c) are attributed to methacrylate of the LBC. The spectra of PEG and GMMA also shows the signals of ethylene oxide portions^{22, 31} at the range of 3.50-4.19 ppm (peak d, e, f and g). As displayed in Figure 1(b), ¹¹B-NMR spectra were used to further characterize the boron ester structure of LBC. TMB exhibits a sharp boron peak at 18.50 ppm, where single broad peak at 23.65 ppm is assigned to the tricoordinated boron atom. ¹³C-NMR and FT-IR spectroscopy analysis were also employed to verify the successful synthesis of the LBC (Figure S1).

The structure of the TBC was also measured by ¹H-NMR spectroscopy, as demonstrated in Figure 2(a). Peaks a, b and e are attributed to the protons of the methacrylate, and peaks c and d are assigned to EO chains. The existence of protons in the EO moiety nearby borate (B-O-CH₂-) was confirmed by the signals in the range of 3.88-4.05 ppm (peak f), which also verified the introduction of borates into the PEGMA. Apparently, when the hydroxyl (-OH) of PEGMA are substituted by the borates, the proton peaks of EO chain-end shifted to downfield due to electron deficient nature of the borates. The TBC was further characterized using ¹¹B-NMR analysis (Figure 2(b)). TMB shows a sharp boron peak at 18.47 ppm, whereas the TBC shows a broad peak at 19.11 ppm in accordance with the characteristic signal of the tricoordinated boron element. The slight peak shift and broadening of the peaks are attributed to the connected EO repeats. ¹³C-NMR and FT-IR spectroscopy analysis were also employed to verify the successful synthesis of the TBC (Figure S2).

The morphology of 3D-BGPEs was characterized by SEM. As exhibited in Figure 3(a), cellulose membrane is comprised of random nanofibers with large pores. Figure 3(b) indicates that a smooth and homogeneous membrane is obtained after polymerization, and the cross-sectional SEM image, as shown in the inset of Figure 3(b), it reveals that the thickness of the 3D-BGPE is about 100 μm. The XRD patterns of PEO and 3D-BGPE are exhibited in Figure 3(c). The sharp peaks at the 19.16°





Scheme 2. (a) The synthesis of the 3D-BGPE membrane via in-situ polymerization, (b) the changes in the Li electrodes with liquid electrolyte and (c) the 3D-BGPE during the Li plating/stripping.

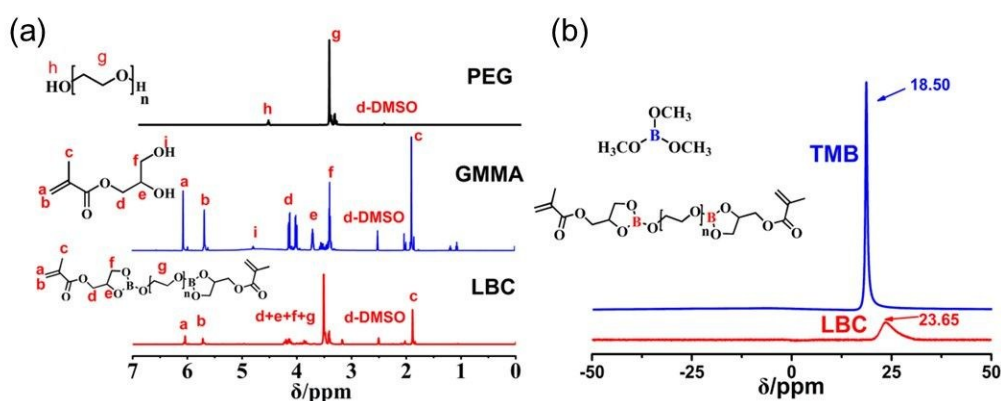


Figure 1. (a) ¹H-NMR spectra of the LBC, GMMA, and PEG. (b) ¹¹B-NMR spectra of the TMB and LBC.

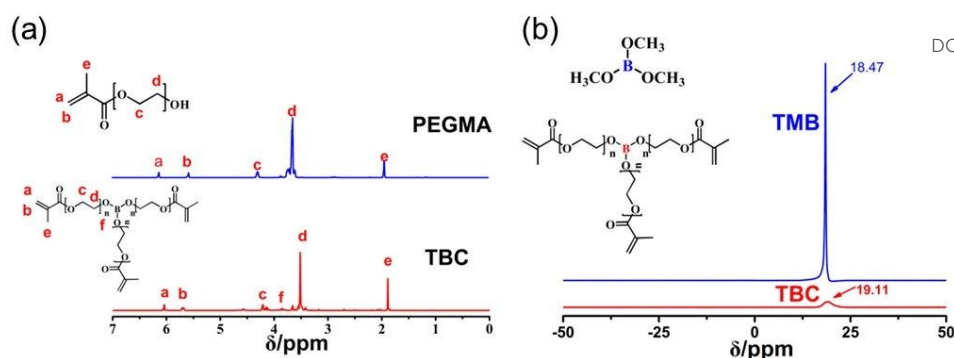


Figure 2. (a) ^1H -NMR spectra of the TBC and PEGMA. (b) ^{11}B -NMR spectra of the TMB and TBC.

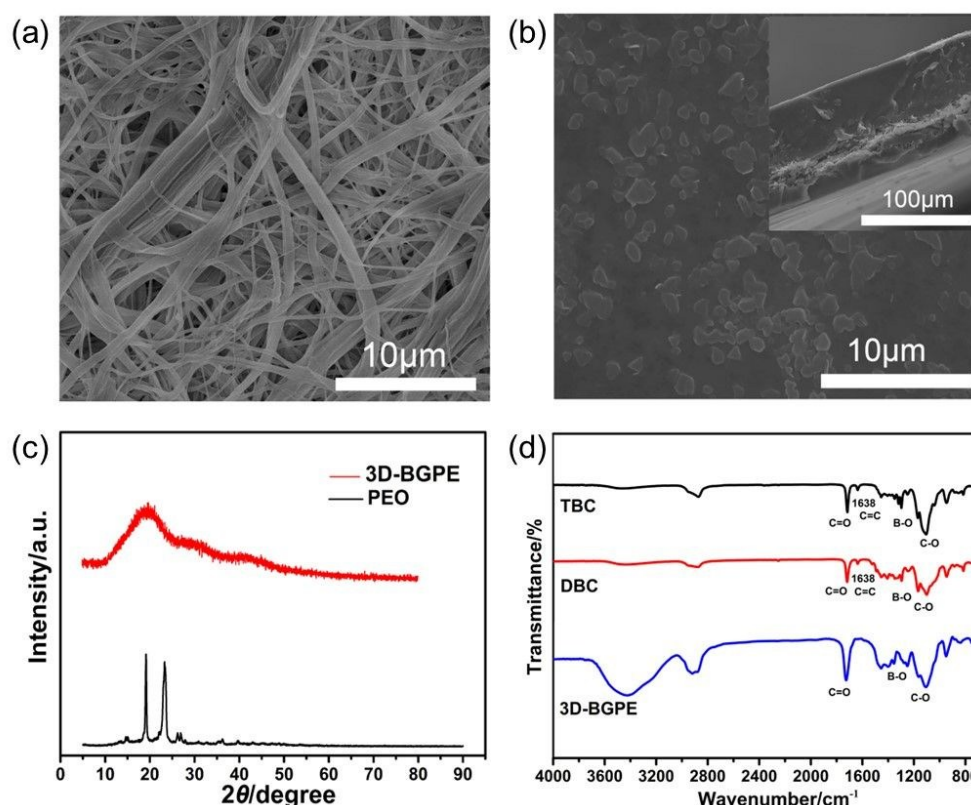


Figure 3. (a) SEM secondary electron image of the pristine cellulose membrane; (b) SEM secondary electron image of the 3D-BGPE; (c) XRD patterns of PEO and 3D-BGPE; (d) FTIR spectra of the LBC, TBC and 3D-BGPE.

and 23.33° showed in the XRD pattern indicates the semi-crystalline structure of the PEO, which is adverse to ion transport and result in lower the ionic conductivity.⁶ In contrast, the cross-linked network of the 3D-BGPE results in distinct decreased crystallinity degree and no obvious peak of the PEO can be obtained, suggesting the amorphous nature of the 3D-BGPE. This is favorable to the Li^+ transfer and ionic conductivity of the 3D-BGPE. The thermal stability was identified via thermogravimetric analysis (Figure S3). It is evident that the 3D-BGPE shows insignificant degradation until temperature achieves to 235°C . The nonlinear changing correlation of weight and temperature suggests nonuniform reaction of cross-linkers and the generation of 3D network.

Thermal behavior was experimented to confirm the safety of the 3D-BGPE. As shown in Figure S4, commercial separator experiences clearly thermal shrinkage after heating at 150°C for 10 min, while the 3D-BGPE shows marginal shape changes under the same condition. These results indicate that the 3D-BGPE can efficiently block the safety issues caused by short circuit and promote the applications at high temperatures. FT-IR analysis was conducted to investigate the polymerization of two monomers (Figure 3(d)). The signals in the range of $1600\text{--}1690\text{ cm}^{-1}$ are generally attributed to C=C stretching of the methacrylate groups³². Moreover, the peaks disappeared after polyreaction compared with the reactants, confirming the complete polymerization reaction of cross-linkers. The

optical photographs shown in Figure S5 demonstrate the gelation evolution where the original flowable liquid precursor solution turns into a uniform gel polymer electrolyte with desirable mechanical strength (Figure S6).

3.2 Electrochemical Properties

Influence of the mole ratios of two monomers on ionic conductivity of 3D-BGPE was investigated. It can be intuitively distinguished from Figure 4(a) that the ionic conductivity of the polymer electrolyte increases with increasing the amount of LBC and achieves a maximum value of $8.4 \times 10^{-4} \text{ S/cm}$ (LBC:TBC = 10:1), and then decreases with further increase of the LBC. Generally, the ionic conductivity increases with increasing the amount of liquid electrolyte in a gel polymer electrolyte (Figure S7), and it is slightly lower than that of liquid electrolyte ($1.1 \times 10^{-3} \text{ S/cm}$ at 30°C). To ensure both high ionic conductivity and reasonably mechanical integrity, the 3D-BGPE with the LBC: TBC ratio of 10:1 and 80% liquid electrolyte was chosen as the research system. The activation energy (E_a) of Li diffusion in the 3D-BGPE was calculated to be $5.30 \times 10^{-2} \text{ eV}$, as demonstrated in Figure 4(b), which is slightly higher than that of liquid electrolyte with commercial membrane

($4.0 \times 10^{-2} \text{ eV}$). As shown in Figure 4(c), stripping of lithium appeared at 0.50 V vs. Li^+/Li and deposition of lithium is formed at -0.48 V vs. Li^+/Li . This process is reversible and there is no apparent oxidation peak until 4.52 V vs. Li^+/Li in the 3D-BGPE, which is slightly higher than that of the liquid electrolyte at 4.48 V (Figure S10). The higher electrochemical window and suppressed anodic current can be attributed to the strong interaction between the boron moieties and anions of the electrolyte salts. This reveals that the 3D-BGPE could be useful in high voltage LMBs.

As an important property of electrolytes, a high t_{Li^+} could effectively reduce concentration polarization and improve the rate performance of batteries.³³ The t_{Li^+} of the 3D-BGPE reaches up to 0.76 (Figure 4(d)), which significantly exceeds that of most commercial liquid electrolytes (0.2~0.4),³⁴ as demonstrated in (Figure S9). This may be attributed to the anion immobilizing behavior of Lewis acidic boron ester. In order to elucidate the underlying reasons for the extremely high t_{Li^+} of 3D-BGPE, the interaction energies between the LBC and TBC molecules and TFSI^- anions were calculated, respectively. As depicted in Figure 4(e-f), the

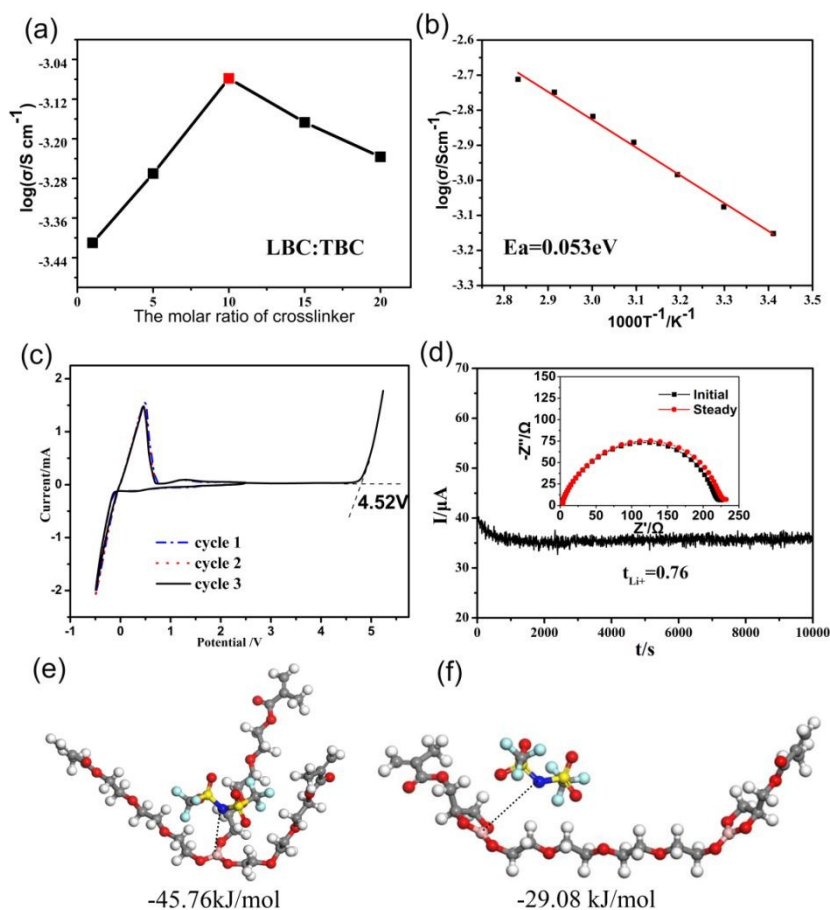


Figure 4. (a) Ionic conductivity dependence on the molar ratio of cross-linkers (LBC:TBC); (b) Arrhenius fitting of ionic conductivity at temperatures from 10 to 70°C ; (c) Cyclic voltammogram of the 3D-BGPE; (d) Current variation with polarization of a Li/3D-BGPE/Li cell with an potential of 10 mV and AC impedance spectra before and after polarization; Schematic illustration of interaction energy induced by (e) the TBC and (f) the LBC with TFSI^- , respectively.

adsorption energies for the TFSI⁻ anion on the LBC and TBC molecules are -29.08 kJ/mol and -45.76 kJ/mol, respectively, indicating that the boron-containing monomers in the 3D-BGPE interact with TFSI⁻ anions strongly. Such adsorption interaction would limit the movement of lithium salt anion (TFSI⁻) and thus raise the t_{Li^+} of the 3D-BGPE substantially. Taking the overall ionic conductivity (σ_{Overall}) and t_{Li^+} in account, it is worth noting that the contribution from Li⁺ conduction ($\sigma_{\text{Li}^+} = (\sigma_{\text{Overall}} \times t_{\text{Li}^+})$) in the liquid electrolyte and 3D-BGPE electrolyte are 3.9×10^{-4} S/cm and 6.3×10^{-4} S/cm, respectively. This suggests that the in-situ formed 3D-BGPE electrolyte may substantially facilitate Li⁺ transport and Li uniform deposition when compared with the liquid electrolyte.

3.3 Battery Performance

The interface compatibility plays a key role in the cycling performance of LMBs, which could be measured by surveying the time evolution of the impedance spectra of a symmetric cell. As shown in Figure 5(a), the interfacial resistance between lithium electrodes and the 3D-BGPE increased from 221 Ω to 271 Ω within 8 days and became stable subsequently. Galvanostatic polarization test of the Li/3D-BGPE/Li cells was carried out to investigate the dynamic stability of the interface between lithium electrodes and the 3D-BGPE (Figure 5(b)). The cells were charged for 1 h and discharged for 1 h with a steady current density of 0.50 mA/cm². The over potential for deposition or stripping of Li was about 38 mV at the beginning and varied with time, because of the formation of SEI film on the lithium electrodes. The over potential changed to around 31 mV after 200 cycles and the polarization kept constant in the subsequent 1000 h without short circuit, indicating excellent compatibility between the 3D-BGPE and the lithium electrodes. In contrast, the symmetric cell using liquid electrolyte (LE) show obvious short circuit after 500 h. This indicates that a lithium metal battery employing the 3D-BGPE would have an extended cycle life.

In order to evaluate the electrochemical performances of the prepared 3D-BGPE, LiFePO₄/3D-BGPE/Li cells were assembled through in situ strategy. It is evident from Figure 6(a) that the increase of TBC effectively improves the capacity of batteries with larger current densities, while the LBC is more favorable

with smaller current densities. Among these, the 3D-BGPE with LBC:TBC = 10:1 exhibits the best overall performances at various current densities. The linear chains of LBC mainly accelerate the Li⁺ transference, while the branched TBC mainly playing as a cross-linking structure and building 3D networks. The charge-discharge profiles with this LBC:TBC ratio, as exhibited in Figure 6(b), deliver reversible capacities of 118.2, 130.1, 137.4 and 145.8 mAh/g at 1, 0.5, 0.2 and 0.1C, respectively. The long-term cycling performance of LiFePO₄/3D-BGPE/Li cell and LiFePO₄/LE/Li cell are compared in Figure 6(c). After 200 cycles, the LiFePO₄/LE/Li cell appears reduced capacity and Coulombic efficiency, which may be attributed to the efficiency loss of Li anodes. In comparison, the LiFePO₄/3D-BGPE/Li cell exhibits no obvious fading either in capacity or in Coulombic efficiency, with a capacity retention of 89.73% after 400 cycles and a stable Coulombic efficiency, indicating improved interfacial stability between electrodes and 3D-BGPE. Even at a higher current density of 2C, the LiFePO₄/3D-BGPE/Li cell exhibits a reasonably high reversible capacity (~105 mAh/g) and excellent cyclability, as shown in Figure S11.

SEM analysis was performed to observe the differences in surface morphology of lithium anodes after 10 cycles and 50 cycles at 0.5 C. Lithium metal anodes after cycling with the liquid electrolyte and the 3D-BGPE are compared to show the difference in dendrite growth. As shown in Figure 7(a, b), smooth and dense morphology without cracks can be achieved using the 3D-BGPE. Little lithium dendrites present on the anode surface. On the contrary, some dendrites could be easily observed on the lithium electrode surface with the liquid electrolyte after cycling at 0.5 C for 10 cycles (Figure 7(d)), and massive sharp dendrites are easily obtained on the lithium electrode surface after 50 cycles at 0.5 C (Figure 7(e)). The corresponding EIS results, as shown in Figure 7(c, f) and Table S1, further verify the more stable electrode/3D-BGPE interfaces upon cycling when compared with the LE counterparts. These phenomena reveal the function of the 3D-BGPE on blocking the growth of lithium dendrites compared to the conventional liquid electrolyte.

X-ray photoelectron spectroscopy (XPS) was carried out to affirm the chemical state and composition of the SEIs. The F 1s

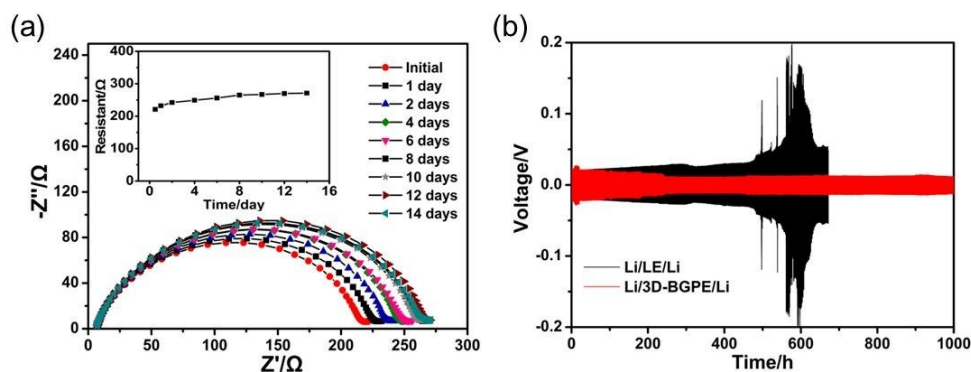


Figure 5. (a) The AC impedance of symmetric Li/3D-BGPE/Li cell at various aging times; (b) Galvanostatic Li plating/stripping profiles of Li/3D-BGPE/Li cell and Li/LE/Li cell at current density of 0.5 mA/cm².

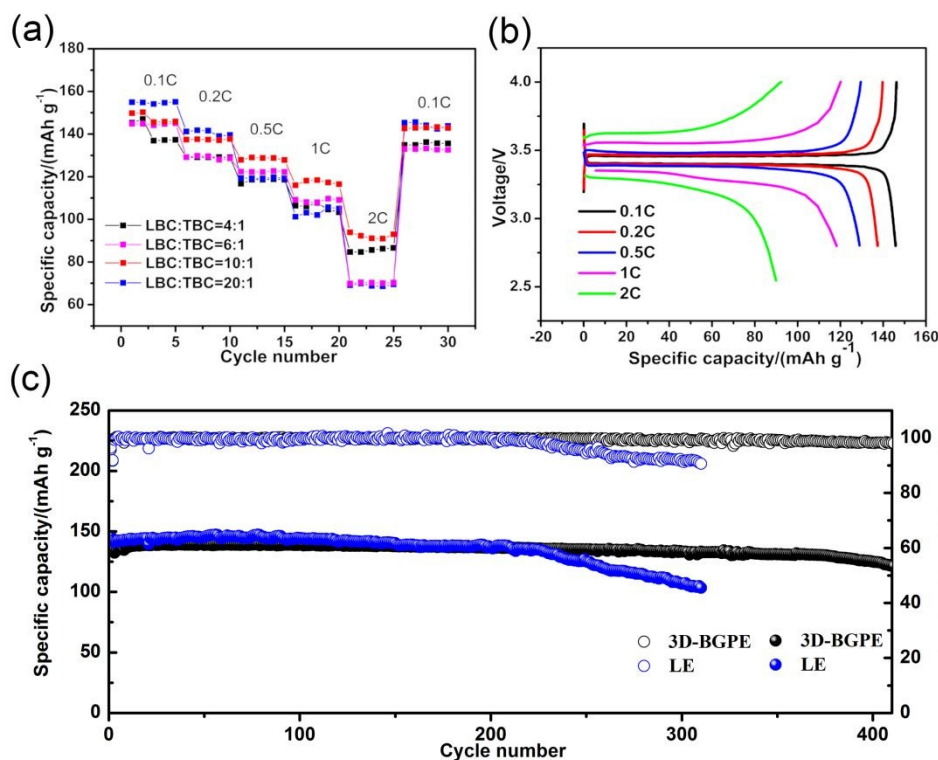


Figure 6. (a) Rate capacities of LiFePO₄/3D-BGPE/Li; (b) Charge and discharge curves at various rates; (c) Cycling performance of LiFePO₄/3D-BGPE/Li cell and LiFePO₄/LE/Li cell at 0.5 C at 30 °C.

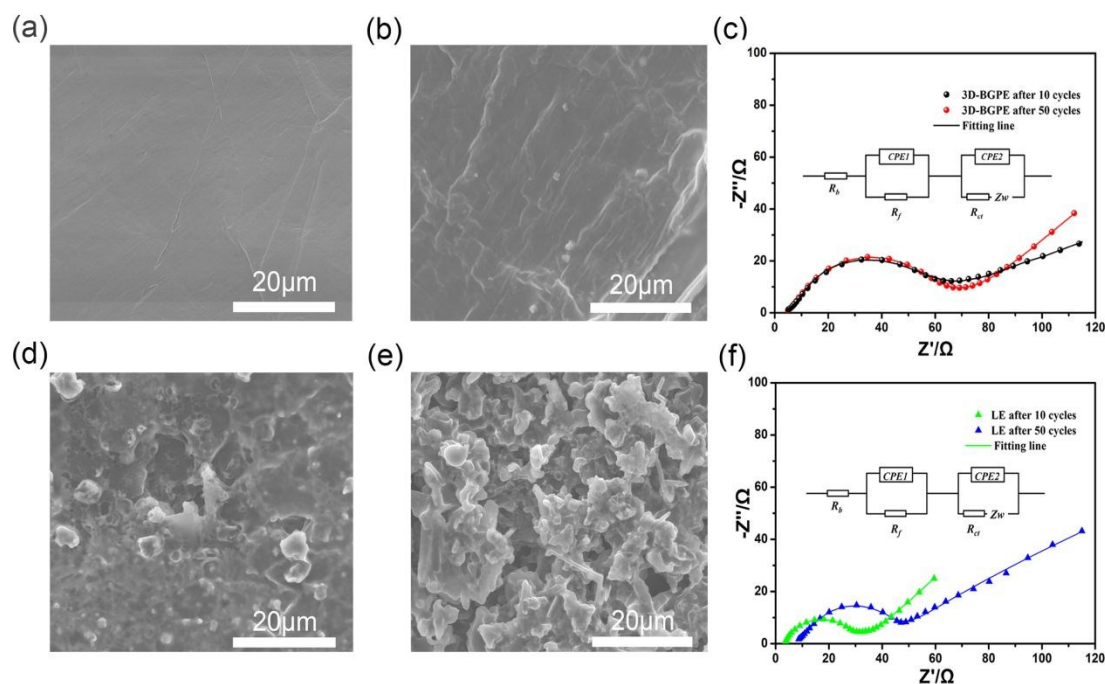


Figure 7. SEM images and corresponding EIS spectra of Li metal electrodes taken from (a-c) LiFePO₄/3D-BGPE/Li cell and (d-f) LiFePO₄/LE/Li cell after 10 and 50 cycles at 0.5 C.

and C 1s XPS spectra of the SEI obtained from lithium metal anodes using liquid electrolyte are presented in Figure 8(a, b, c). The characteristic peaks at 686.3 and 688.5 eV in Figure 8(a) are assigned to the C-F and LiF, respectively. It is apparent that

no significant change occurs during electrochemical cycling. As for the C 1s spectra, as compared in Figure 8(b), the signals of COR, COOR, CO₃²⁻, CF₃, C-C, locate at 287 eV, 289 eV, 290 eV, 293 eV, 284.8 eV, respectively. The amount of COR group

increases after cycling, indicating the continuous decomposition of carbonate solvents. For the 3D-BGPE containing system, as shown in Figure 8(d, e, f), an extra B-O peak at 192.3 eV can be observed, suggesting that boron containing motifs contribute to the build of the SEI film. Interestingly, the major components of the SEI layer are the COR and C-C groups, while the CF_3 component is markedly

suppressed when compared with the LE counterpart. These imply that the formation of a thin SEI layer mainly constituted by cross-linked organic species and reduces the decomposition of lithium salts in the electrolytes, and thus increases the ionic conductivity and improves cell performance.

Table 1 shows the electrochemical performances of various gel

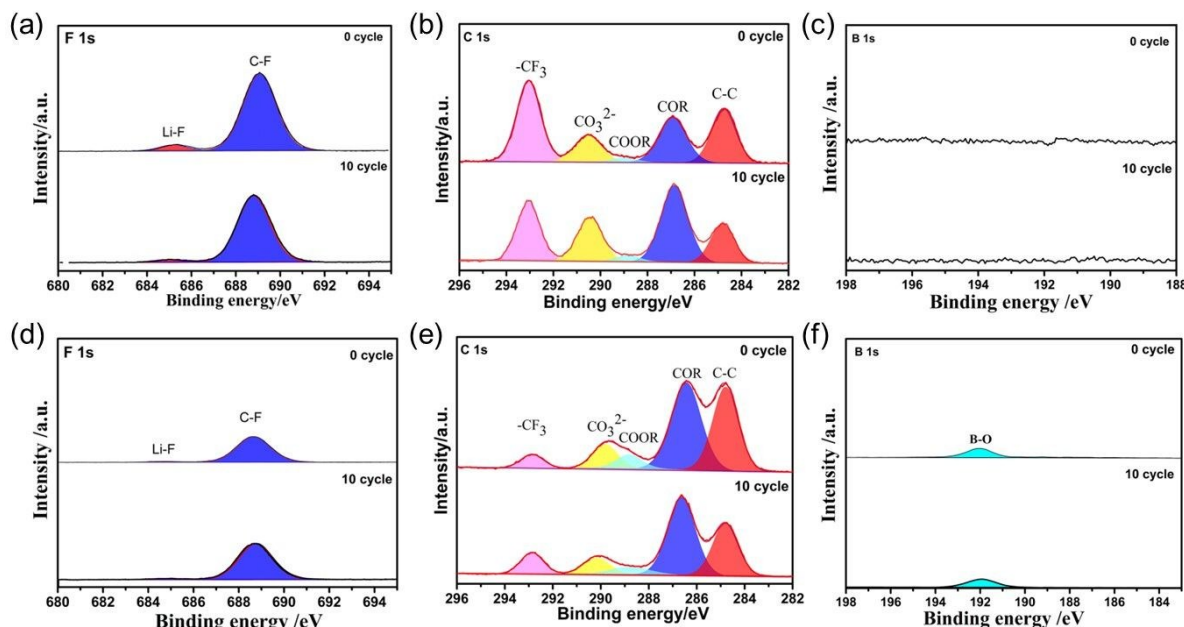


Figure 8. XPS characterization of the SEI components produced on cycled lithium anode surface. (a-c) F1s, C1s and B 1s XPS spectra of for lithium anodes using liquid electrolyte; (d-f) F1s, C1s and B 1s XPS spectra taken from lithium anodes using the 3D-BGPE.

Table 1 Electrochemical performances of various gel polymer electrolytes reported in the literature

GPE ingredients	Room temperature ionic conductivity/(Scm^{-1})	Li^+ transference number	Preparation method	Cell(cycles)capacity retention /C-rate	Ref.
Poly(vinylene carbonate)	5.59×10^{-4}	0.34	In-situ	LFMP/Li(100th)95.4%/1C	35
Interpenetrating network of poly(diethylene glycol carbonate)	1.64×10^{-4}	0.47	In-situ	LFP/Li(100th)95%/1C	29
Borate esters with silica articles	8.0×10^{-4}	0.34-0.65	Ex-situ	-	36
Zwitterionic polymer networks in ionic liquids	7×10^{-4}	0.6	Ex-situ	NCM523 /graphite (significant capacity fade after 50th)/0.5C	37
2D boron nitride nanoflakes in PVDF	8.1×10^{-4}	0.62	Ex-situ	LFP/Li(300th)/88%/1C	38
Silyl-terminated 3D polymer electrolyte	3.6×10^{-4}	0.65	Ex-situ	LFP/Li(600th)/99.7%/3C	39
Interpenetrating network of poly(ether-crylate)	2.2×10^{-4}	0.65	Ex-situ	LFP/Li(200th)85%/1C	40
Polyethylene -supported gel polymer electrolyte	4.5×10^{-4}	0.72	Ex-situ	LCO/Li(300th)62.7%/0.2C	41
A dual-salt with 3D cross-linked polymer network	5.6×10^{-4}	0.72	In-situ	LFP/Li(300th)87.9%/0.5C	42
A borate concentrated 3D cross-linked network	8.4×10^{-4}	0.76	In-situ	LFP/Li(400th)89.7%/0.5C	This work

polymer electrolytes in the last few years. GPEs with high ionic conductivities and Li^+ transference number can reduce the concentration polarization effects and suppress lithium dendrites growth. Several approaches have been reported to promote the ion transport efficiency. It is worth proposing that our work offers higher ionic conductivity and transference number compared to most reports. 3D-BGPE incorporates two kinds of boron moieties into the polymer matrix via in situ polymerization. With a vacant p-orbital, boron moieties work as a Lewis acidic center that interacts with the anion of electrolyte salts to increase the dissociation of ion pairs and improve ionic conductivity and transfer number. In situ-generated 3D-BGPE improves the interfacial compatibility between the lithium anode and electrolyte. Therefore, the 3D-BGPE obtains high ionic conductivity and transfer number, simultaneously construct a stable SEI film on lithium anodes, which is favorable to the inhibition of dendrite and long-term cycling.

Conclusions

In this work, 3D-BGPE rich in boron ester with 3D cross-linked polymer structure has been successfully fabricated by in situ strategy, where the linear chains of LBC greatly accelerate Li^+ transference, while the branched TBC mainly act as a cross-linking structure. The optimized 3D-BGPE demonstrates a prominent ionic conductivity of $8.4 \times 10^{-4} \text{ S/cm}$ at 30°C , a high Li^+ transference number of 0.76, a wide potential window up to 4.52 V and superior mechanical property with tensile strength of 7.6 MPa. The $\text{Li}/3\text{D-BGPE}/\text{Li}$ symmetric cell maintains an extremely stable and low overpotential without short circuiting over 1000 h. The $\text{LiFePO}_4/3\text{D-BGPE}/\text{Li}$ cell achieves 89.73% capacity retention after cycling for 400 cycles.

Conflicts of interest

There are no conflicts to declare.

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